LSIS and Convolution

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Topic: Image Processing I, Module: Imaging

First Principles of Computer Vision

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Linear Shift Invariant System (LSIS)



Study of Linear Shift Invariant Systems (LSIS) leads to useful image processing algorithms.

LSIS: Linearity





LSIS: Shift Invariance

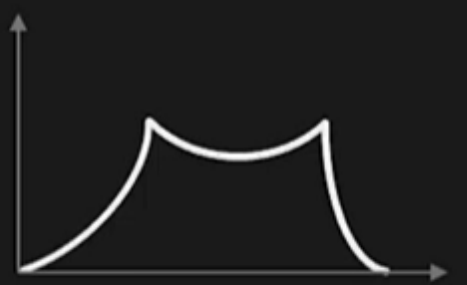


LSIS: Shift Invariance

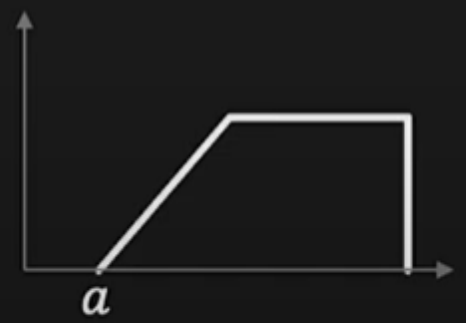


$f(x)$



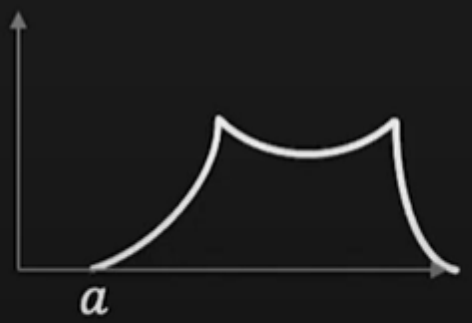


$g(x)$



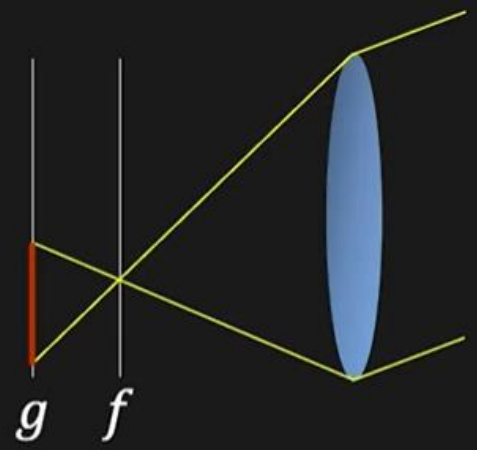
$f(x-a)$





$g(x-a)$

Ideal Lens is an LSIS



Defocused Image (g): Processed version of Focused Image (f)

Linearity: Brightness variation

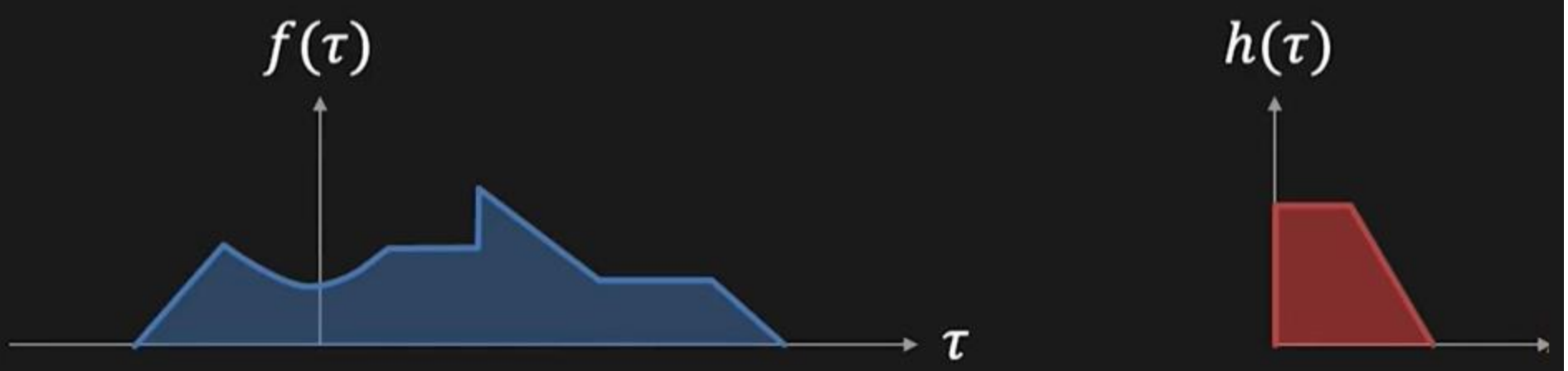
Shift invariance: Scene movement

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Convolution

Convolution of two functions $f(x)$ and $h(x)$

$$g(x) = f(x) * h(x) = \int_{-\infty}^{\infty} f(\tau)h(x-\tau)\,d\tau$$



Convolution

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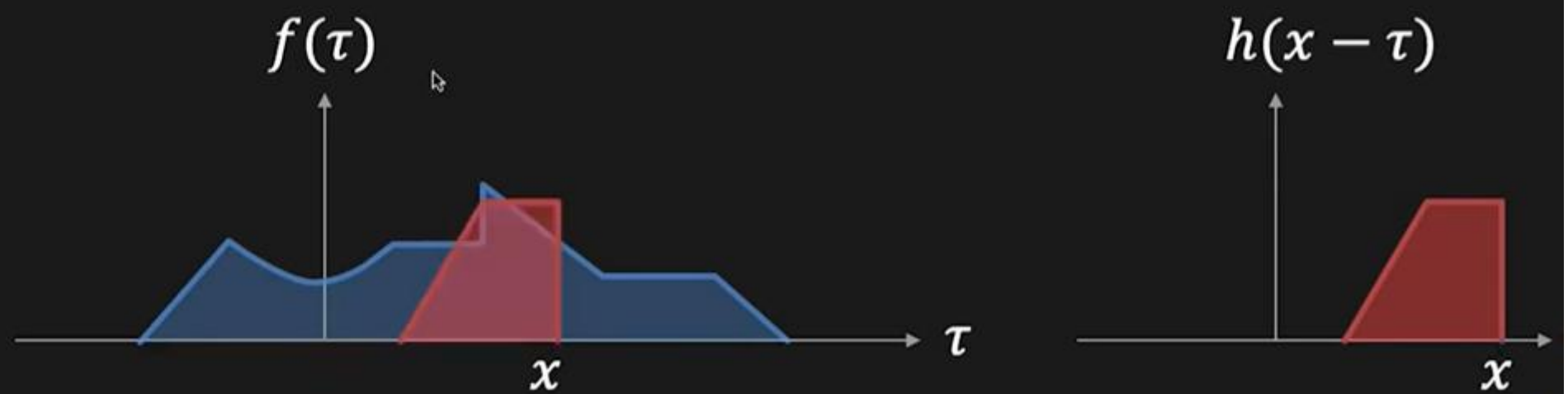
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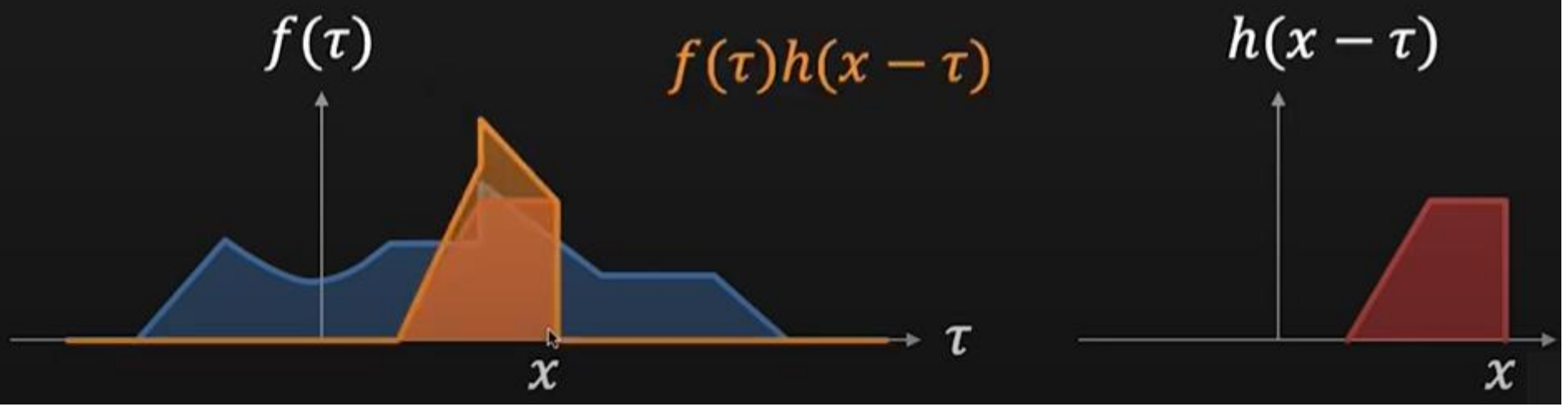
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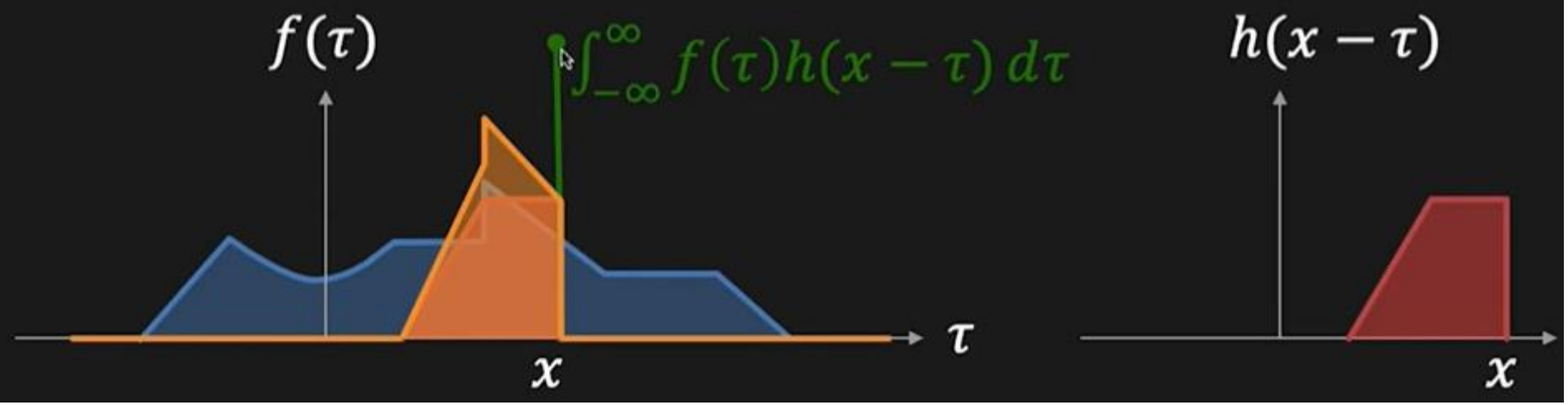
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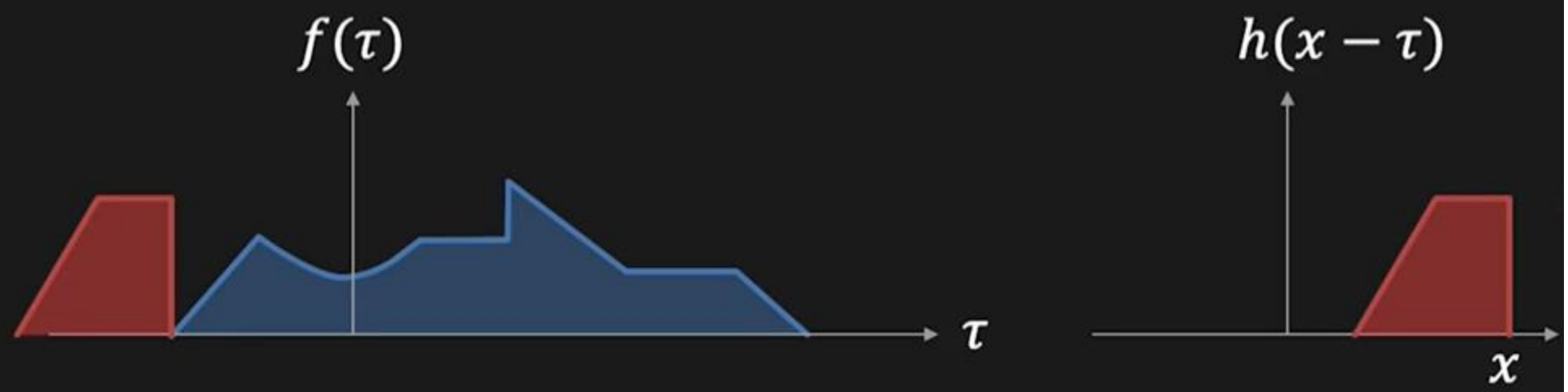
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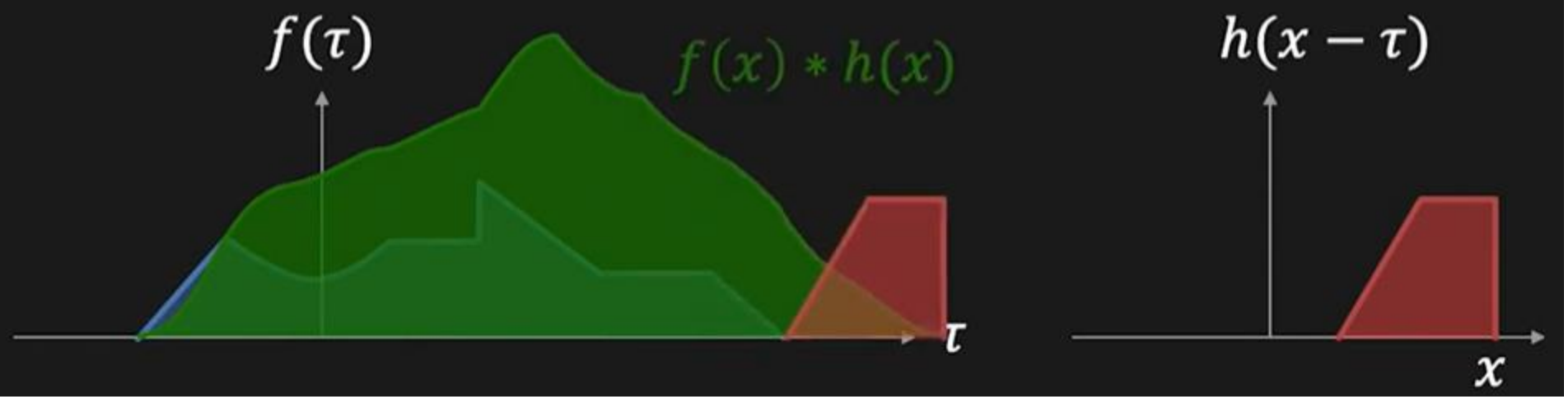
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Convolution

Convolution of two functions $f(x)$ and $h(x)$

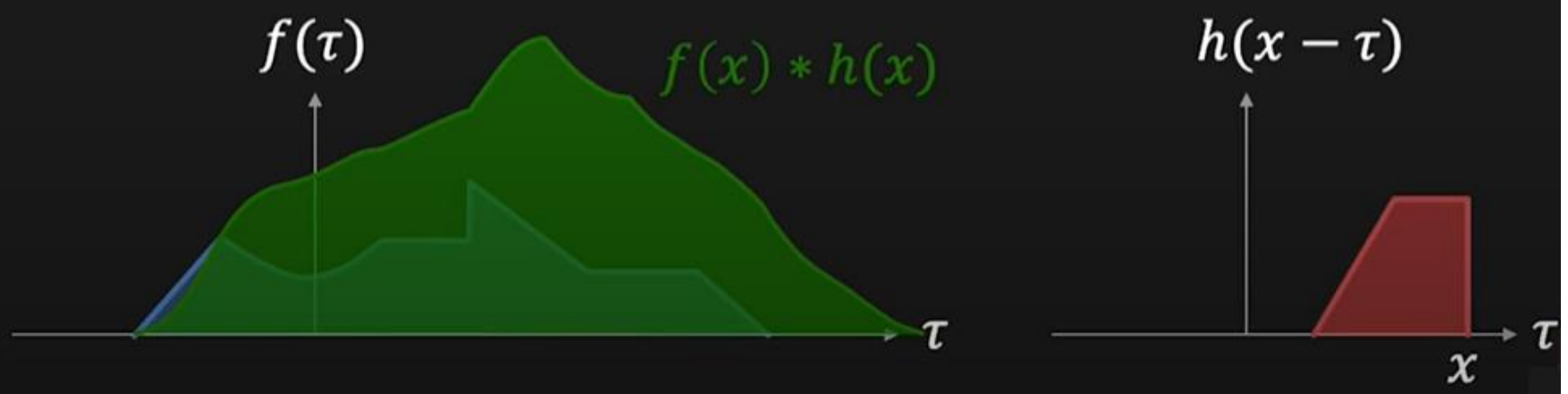
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Convolution

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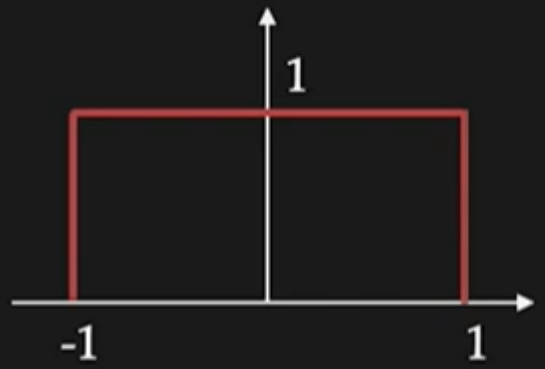


LSIS implies Convolution and Convolution implies LSIS

Convolution: Example

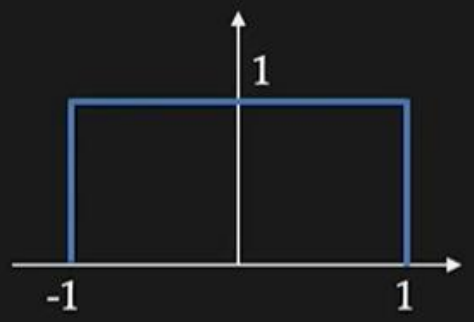


$f(x)$

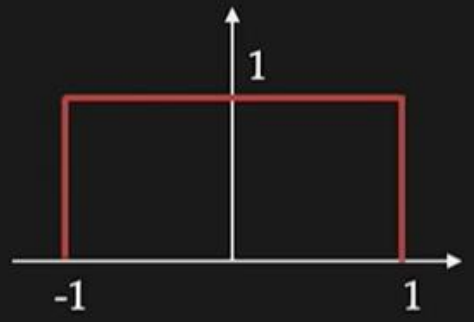


$h(x)$

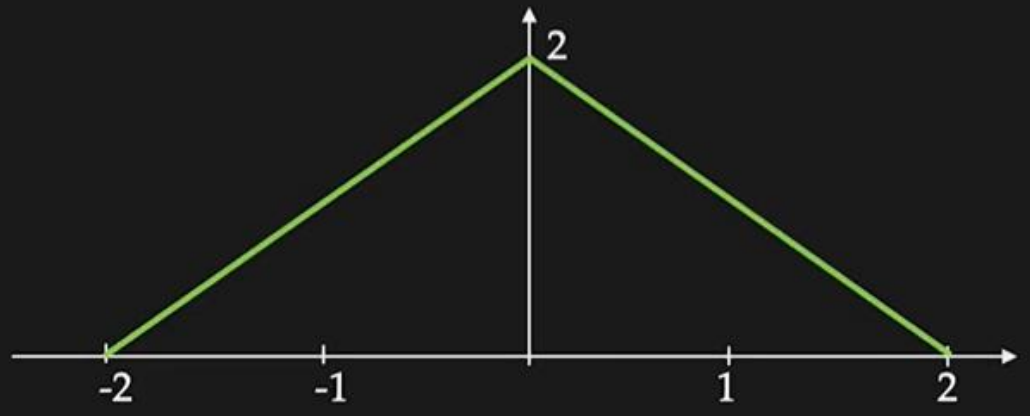
Convolution: Example



$f(x)$



$h(x)$



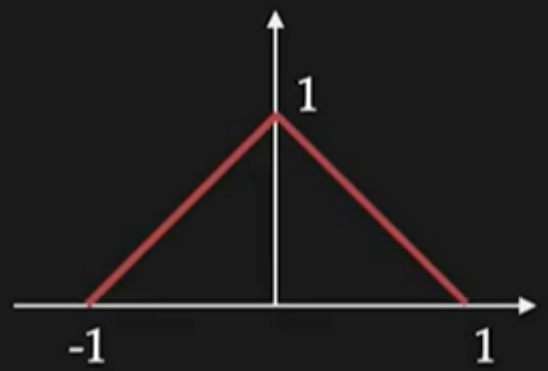
$f(x) * h(x)$

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Convolution: Example



$f(x)$

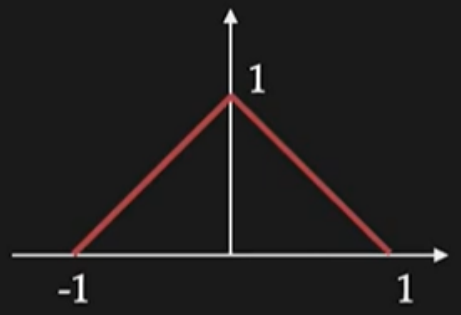


$h(x)$

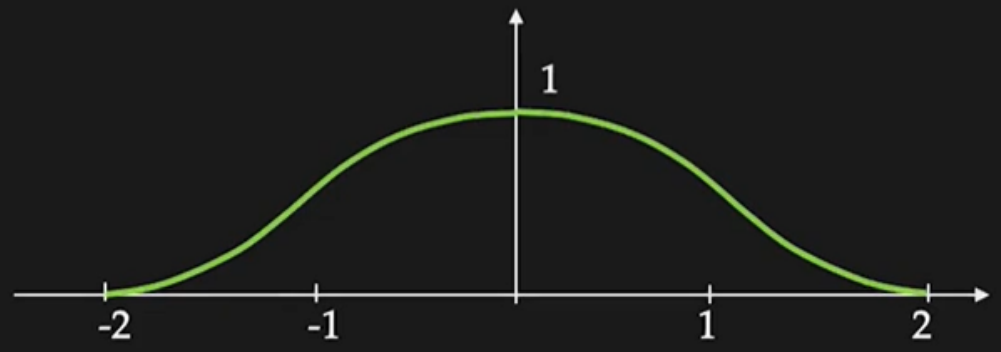
Convolution: Example



$f(x)$



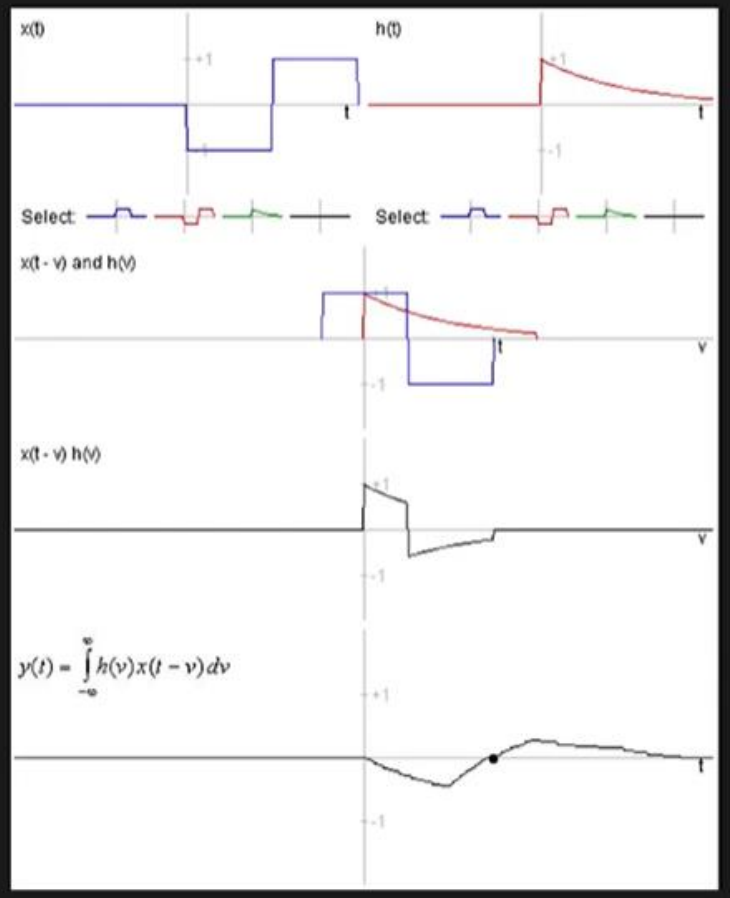
$h(x)$



$f(x) * h(x)$

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Convolution: Online Demo



http://www.jhu.edu/signals/convolve/

Convolution is LSIS

Linearity:

Let: $g_1(x) = \int_{-\infty}^{\infty} f_1(\tau)h(x-\tau)\,d\tau$ and $g_2(x) = \int_{-\infty}^{\infty} f_2(\tau)h(x-\tau)\,d\tau$

Then:

$$\int_{-\infty}^{\infty} (\alpha f_1(\tau) + \beta f_2(\tau))h(x-\tau)\,d\tau$$

$$= \alpha \int_{-\infty}^{\infty} f_1(\tau)h(x-\tau)\,d\tau + \beta \int_{-\infty}^{\infty} f_2(\tau)h(x-\tau)\,d\tau$$

$$= \alpha g_1(x) + \beta g_2(x)$$

Convolution is LSIS

Shift Invariance:

Let: $g(x) = \int_{-\infty}^{\infty} f(\tau)h(x-\tau)\,d\tau$

Then:

$$\int_{-\infty}^{\infty} f(\tau - a)h(x-\tau)\,d\tau$$

$$= \int_{-\infty}^{\infty} f(\mu)h(x-a-\mu)\,d\mu \qquad \text{(Substituting } \mu = \tau - a\text{)}$$

$$= g(x-a)$$

Can we find h?



$$g(x) = \int_{-\infty}^{\infty} f(\tau)h(x-\tau)\,d\tau$$

Can we find h?



$$g(x) = \int_{-\infty}^{\infty} f(\tau)h(x-\tau)\,d\tau$$

What input f will produce output $g = h$?

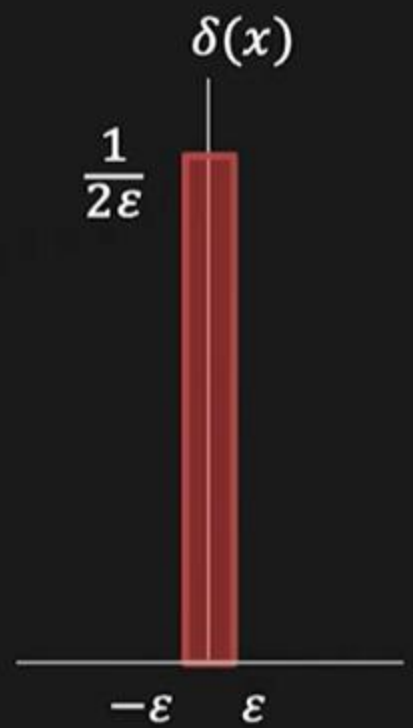
$$h(x) = \int_{-\infty}^{\infty} ?\,(\tau)h(x-\tau)\,d\tau$$

Unit Impulse Function

$$\delta(x) = \begin{cases} 1/2\varepsilon, & |x| \le \varepsilon \\ 0, & |x| > \varepsilon \end{cases}$$

$$\varepsilon \to 0$$

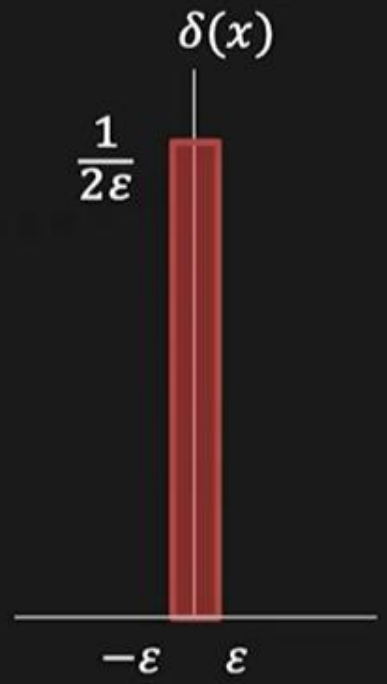
$$\int_{-\infty}^{\infty} \delta(\tau)\, d\tau = \frac{1}{2\varepsilon} . 2\varepsilon = 1$$



Unit Impulse Function

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$$\varepsilon \to 0$$



$$\int_{-\infty}^{\infty} \delta(\tau)\, d\tau = \frac{1}{2\varepsilon}.2\varepsilon = 1$$

$$\int_{-\infty}^{\infty} \delta(\tau) b(x-\tau)\, d\tau = b(x)$$

Sifting Property

Impulse Response



$$g(x) = f(x) * h(x)$$

$$g(x) = \int_{-\infty}^{\infty} f(\tau)h(x-\tau)\,d\tau$$



Unit Impulse

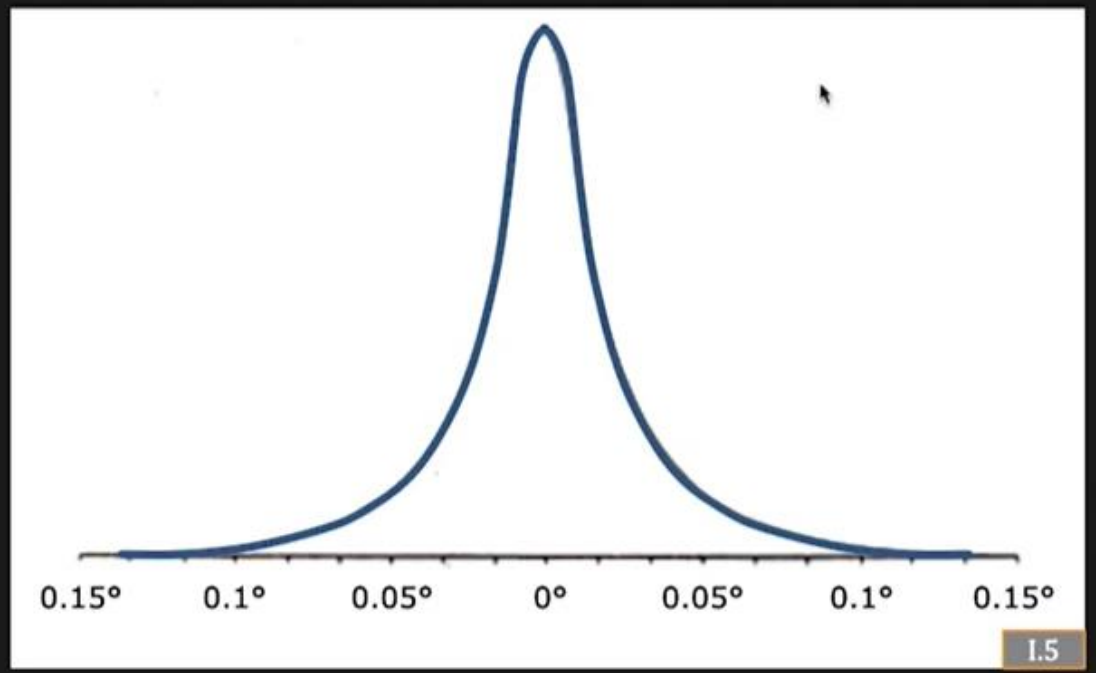
Impulse Response

$$h(x) = \delta(x) * h(x)$$

$$h(x) = \int_{-\infty}^{\infty} \delta(\tau)h(x-\tau)\,d\tau$$

Impulse Response of Human Eye

Distant Star $\delta(x, y)$ → Human Eye → Point Spread Function (PSF) $h(x, y)$



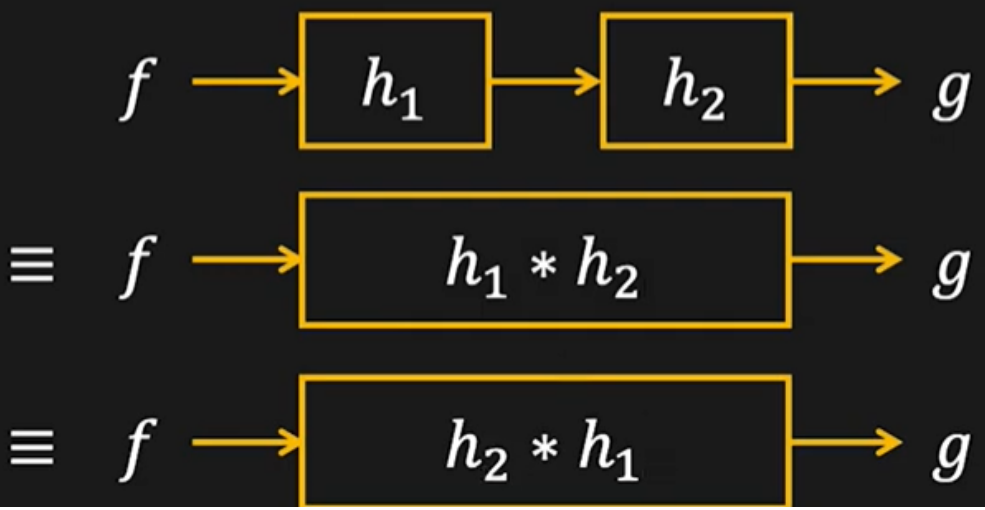
Human Eye PSF

Properties of Convolution

Commutative $a * b = b * a$

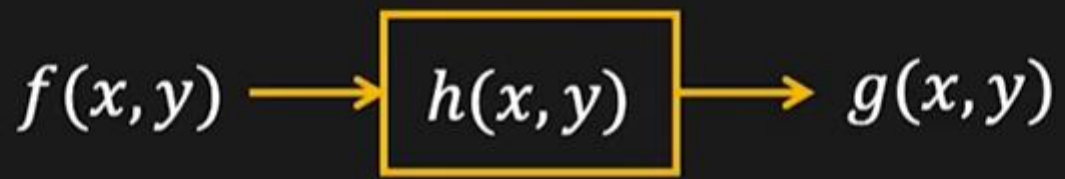
Associative $(a * b) * c = a * (b * c)$

Cascaded System



2D Convolution

LSIS:



Convolution:

$$g(x,y) = \iint_{-\infty}^{\infty} f(\tau,\mu)h(x-\tau,y-\mu)\,d\tau d\mu$$